MTH441: Lab 6

Model Selection

1. Load the `ISLR` library in R and study the `Auto` dataset and create a scatterplot of the dataset, where miles per gallon, `mpg` is the response.
   i. Which columns are categorical/factor variables?
   ii. Convert factor variables to factors. For example:

   ```
   Auto$origin <- as.factor(Auto$origin)
   ```

   iii. Color the point of the scatterplot using `origin` as a factor. Visually, can you say if `origin` will be significant in explaining something about `mpg`?
   iv. Visually, do you think an interaction term with other covariates is necessary?
2. Fit an appropriate full model for the response `mpg`. Calculate the AIC value using the formula. You may use `model.matrix` function.
3. Use `stepAIC` function in `MASS` library to find the best reduced model.
4. Repeat 2. and 3. the same for the `Boston` dataset from before.
5. Repeat 2. and 3. for the Titanic dataset and Logistic Regression.